

School: SET		Batch: 2023-27	
Program: B-TECH		Current Academic Year: 2023-24	
Branch: CSE		Semester:	
1	Course Code	CSA303	
2	Course Title	Deep Learning and its Applications	
3	Credits	3	
4	Contact Hours (L-T-P)	3-0-0	
	Course Status	CORE	
5	Course Objective	This course aims to present the mathematical, statistical and computational challenges of building stable representations for high-dimensional data, such as images, text and data. We will delve into selected topics of Deep Learning, discussing recent models from both supervised and unsupervised learning. Special emphasis will be on convolutional architectures, invariance learning, unsupervised learning and non-convex optimization. To understand and demonstrate how to solve general learning from a large series of data using computer based deep learning algorithms	
6	Course Outcomes (CO's)	On successful completion of this module students will be able to: 1. Recall Neural Networks relate it with Deep Learning concepts to solve real life applications 2. Compare and classify Regularization approaches for Deep Learning. 3. Build Convolutional Neural Networks models for image analysis and its applicability in societal problem solving. 4. Examine the Sequence models and analyse the relationships among them. 5. Assess the different Deep learning models based on their design processes. 6. Predict the behaviour of Deep learning models and apply them to solve real life applications.	
7	Course Description	This course starts with introduction to Deep Learning and further build, train, and deploy real world applications such as object recognition and Computer Vision, image and video processing, text analytics, Natural Language Processing, recommender systems, and other types of classifiers.	
8	Syllabus Outline		CO Mapping
	Unit 1	Deep Feed forward Networks	
	A	Recall Neural networks, Deep learning and its Practical aspects for real life applications ,Introduction to Simple Deep Neural Networks, Platform for Deep Learning, Deep Learning Software Libraries	CO1, CO6

	B	Introduction to Deep Feed Forward Networks ,Learning XOR, Gradient-Based Learning, Activation Functions, ReLU, Softmax, Sigmoid , Error Functions	CO1
	C	Architecture Design- Hidden Units Back-Propagation and Other Differentiation Algorithms	CO1
	Unit 2	Regularization for Deep Learning	
	A	Parameter Norm Penalties, Norm Penalties as Constrained Optimization, Regularization and Under-Constrained Problems, Dataset Augmentation, Noise Robustness, Semi-Supervised Learning, Multitask Learning, Early Stopping, Parameter Tying and Parameter Sharing, Bagging , Drop Out, Difficulty of training deep neural networks, Greedy layer wise training, Adversarial Training	CO2
	B	How Learning Differs from Pure Optimization, Challenges in Neural Network Optimization, Basic Algorithms: Stochastic Gradient Descent, Momentum, Nesterov Momentum Parameter Initialization Strategies Algorithms with Adaptive Learning Rates, AdaGrad. RMSProp. Adam	CO2
	C	Introduction to Autoencoder, Undercomplete Autoencoder, Regularized Autoencoders, Representational Power, Layer Size and Depth. Stochastic Encoders and Decoders, Applications of Encoder Decoder models	CO2
	Unit 3	Convolutional Neural Networks	
	A	Why CNN?, Its role, significance and applicability in societal problem solving , The Convolution Operation, Motivation, Pooling, The Neuroscientific Basis for Convolutional Networks	CO1, CO3, CO6
	B	Prior probability distribution, Convolution and Pooling as an Infinitely Strong Prior, Variants of the Basic Convolution Function, Structured Outputs, Data types with different dimensionalities and number of channel	CO1, CO3
	C	Efficient Convolution Algorithms, Random or Unsupervised Features of CNN , Normalization, Applications of CNN in Computer Vision – ImageNet	CO1, CO3, CO6
	Unit 4	Sequence Modeling: Recurrent Neural Networks	
	A	Sequence Learning Problems , Recurrent Neural Network and its significance in real world, RNN model, Backpropagation through time ,Bidirectional RNNs	CO4, CO6
	B	Different types of RNNs, Gated Recurrent Unit (GRU) Recursive Neural Networks , The Challenge of Long-Term Dependencies	CO4
	C	Introduction of Long Short Term Memory Neural Networks, Learning Algorithm of LSTM/ RNN Bidirectional LSTMs	CO4
	Unit 5	Deep Networks and design process	

	A	Introduction to Generative Adversarial Networks , Generative Adversarial Networks – Architecture,			CO5,CO6
	B	DCGAN, GAN hack, Applications of Generative Adversarial Networks			CO5,CO6
	C	Practical design process for deep learning techniques based on real world problems: Performance Metrics , Default Baseline Models, Determining Whether to Gather More Data, Selecting Hyperparameters, Debugging Strategies			CO5,CO6
	Mode of examination	Theory			
	Weightage Distribution	CA	MTE	ETE	
		25%	25%	50%	
	Text Books	1. Deep Learning, by Goodfellow I., Bengio Y. & Courville A. (2016) 2. Visualizing and Understanding Convolutional Networks, by Matt Zeiler, Rob Fergus 3. TensorFlow: a system for large-scale machine learning, by Martín A., Paul B., Jianmin C., Zhifeng C., Andy D. et al. (2019)			
	Reference Books	1. Deep learning in neural networks, by JuergenSchmidhuber (2015) 2. https://cs230.stanford.edu/syllabus/ 3. https://towardsdatascience.com/september-edition-machine-learning-case-studies-a3a61dc94f23 4. Deep Learning: A Practitioner's Approach by Josh Patterson, O'Reilly.			

CO and PO Mapping

S. No.	Course Outcome (CO)	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1	Recall Neural Networks relate it with Deep Learning concepts to solve real life applications	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2	Compare and classify Regularization approaches for Deep Learning.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3	Build Convolutional Neural Networks models for image analysis and its applicability in societal problem solving.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4	Examine the Sequence models and analyse the relationships among them.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5	Assess the different Deep learning models based on their design processes.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6	Predict the behaviour of Deep learning models and apply them to solve real life applications.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

PO and PSO mapping with level of strength for Course Name: Introduction to Deep Learning

(Course Code- CSA302)

School: SET		Batch : 2023-27	
Program: B-TECH		Current Academic Year: 2023-24	
Branch: CSE		Semester:	
1	Course Code	CAL303	
2	Course Title	Deep Learning and its Applications Lab	
3	Credits	1	
4	Contact Hours (L-T-P)	0-0-2	
	Course Status	CORE	
5	Course Objective	This course aims to present the mathematical, statistical and computational challenges of building stable representations for high-dimensional data, such as images, text and data. We will delve into selected topics of Deep Learning, discussing recent models from both supervised and unsupervised learning. Special emphasis will be on convolutional architectures, invariance learning, unsupervised learning and non-convex optimization. To understand and demonstrate how to solve general learning from a large series of data using computer based deep learning algorithms	
6	Course Outcomes (CO's)	On successful completion of this module students will be able to: 1. Define and show the implementation of Deep Learning concepts to solve real life applications 2. Compare and classify Regularization approaches for Deep Learning. 3. Build Convolutional Neural Networks models for image analysis and its applicability in societal problem solving. 4. Examine the Sequence models and analyses the relationships among them. 5. Assess the different Deep learning models based on their design processes. 6. Predict the behavior of Deep learning models and apply them to solve real life applications.	
7	Course Description	This course starts with introduction to Deep Learning and further build, train, and deploy real world applications such as object recognition and Computer Vision, image and video processing, text analytics, Natural Language Processing, recommender systems, and other types of classifiers.	
8	Outline syllabus		CO Mapping
	Unit 1	Deep Feed Forward Networks	
	1	Write a program to implement Deep Feed Forward Network to predict Housing price available on Kaggle.	CO1
	2	Write a program to implement classification using Deep Feed Forward Network on dataset available on Kaggle.	CO1
	Unit 2	Regularization for Deep Learning	

	3	Write a program to implement regularization to overcome over fitting problem in Housing price Prediction	CO2
	Unit 3	Convolutional Neural Networks	
	4	Digit Recognition from MNIST : Given a zip codes hand written on envelops, identify the digit for each hand written character. A model of this problem would allow a computer program to read and understand handwritten zip codes and sort envelops by geographic region.	CO1, CO3, , CO6
	5	Dog-Breed Classifier: Design and train a convolutional neural network to analyze images of dogs and correctly identify their breeds. Use transfer learning and well-known architectures to improve this model—this is excellent preparation for more advanced applications.	CO1, CO3
	Unit 4	Sequence Modelling: Recurrent Neural Networks	
	6	Stock market prediction on NASDAQ stocks: Given the current and past price movements for a stock, determine whether the stock should be bought, held or sold. A model of this decision problem could provide decision support to financial analysts.	CO4, CO6
	7	The Slot-Filling (Spoken Language Understanding) consists in assigning a label to each word given a sentence. It's a classification task.	CO4
	8	Write a program to implement credit card fraud detection prediction	CO4
	Unit 5	Deep Networks and design process	
	9	Implement Convolutional Autoencoders in Python with Keras on MNIST dataset	CO5, CO6
	10	Write a program to implement Generative Adversarial Network	CO5, CO6
	Mode of examination	Practical	
	Weightage Distribution	CA 25%	MTE 25%
			ETE 50%
	Text book/s*	1. Lipschutz, "Data Structures" Schaum's Outline Series, TMH	
	Other References	1. Aaron M. Tenenbaum, Yedidyah Langsam and Moshe J. Augenstein "Data Structures Using C and C++", PHI 2. Horowitz and Sahani, "Fundamentals of Data Structures", Galgotia Publication 3. Jean Paul Trembley and Paul G. Sorenson, "An Introduction to Data Structures with applications", McGraw Hill 4. R. Kruse et al, "Data Structures and Program Design in C", Pearson Education 5. G A V Pai, "Data Structures and Algorithms", TMH	
	Weblink	https://towardsdatascience.com/getting-rich-quick-with-machine-learning-and-stock-market-predictions-696802da94fe	

	https://www.datacamp.com/community/tutorials/autoencoder-keras-tutorial http://deeplearning.net/tutorial/rnnslu.html	
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CO and PO Mapping

S. No.	Course Outcome (CO)	Program Outcomes (PO) & Program Specific Outcomes (PSO)
1	Define and show the implementation of Deep Learning concepts to solve real life applications	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
2	Compare and classify Regularization approaches for Deep Learning.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
3	Build Convolutional Neural Networks models for image analysis and its applicability in societal problem solving.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
4	Examine the Sequence models and analyse the relationships among them.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
5	Assess the different Deep learning models based on their design processes.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3
6	Predict the behaviour of Deep learning models and apply them to solve real life applications.	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10, PSO1,PSO2,PSO3

PO and PSO mapping with level of strength for Course Name: Introduction to Deep Learning

(Course Code- CAL303)

Course Code_ Course Name	CO's	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
Deep Learning and its Applications Lab (Course Code- CAL303)	CO1	3	3	3	3	2	1	1	1	1	3	1	3	3	3	1
	CO2	3	3	3	3	3	3	3	1	3	3	3	3	3	3	3
	CO3	3	3	3	3	3	2	2	1	2	3	3	3	3	3	3
	CO4	3	3	3	3	3	2	2	1	2	3	3	3	3	3	3
	CO5	3	3	3	3	3	3	3	1	3	3	3	3	3	3	3
	CO6	3	3	3	3	3	3	3	2	3	3	3	3	3	3	3

Average of non-zeros entry in following table (should be auto calculated).

Course Code	Course Name	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CAL 303	Deep Learning and its Applications Lab	3.00	3.00	3.00	3.00	2.83	2.33	2.33	1.17	2.33	3.00	2.67	3.00	3.00	3.00	2.67

Total 40.3

Strength of Correlation

1. Addressed to **Slight (Low=1)** extent
2. Addressed to **Moderate (Medium=2)** extent
3. Addressed to **Substantial (High=3)** extent